

Evana Thomson

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EDUCATION

UNIVERSITY OF CALIFORNIA - BERKELEY

Berkeley, CA

B.S. in Electrical Engineering and Computer Science and Minor in Materials Science & Engineering

May 2028

- Coursework: CS61A: The Structure and The Interpretation of Computer Programs, Math 52: Calculus II, CS61B: Data Structures, CS61C: Computer Architecture and Machine Structures, Math 54: Linear Algebra and Differential Equations

WORK EXPERIENCE

ENGINEERING SOLUTIONS AT BERKELEY

Berkeley, CA

Engineering Consultant: QB3 Chemistry Mass Spectrometry Facility

Feb 2026 – May 2026

- Designing PCB subsystem for a miniaturized mass spectrometer, analyzing detector signals, applying electromagnetism concepts and noise reduction techniques while evaluating manufacturing constraints and feasibility for scalable viral detection devices.

Engineering Consultant: Stealth Startup (Under NDA/Patent-Pending Project)

Aug 2025 – Jan 2026

- Building a closed-loop gas flow control system for corrosive environments by integrating industrial sensors, PID, PCBAs, PLCs, and automated control architectures to achieve precise and reliable system performance.

CALI E-SKIN NEUROPATCH

Berkeley, CA

Lead Microchip Engineer: Stealth Startup (Under NDA)

Nov 2025 – May 2026

- Lead PCB, microchip and sensor system design for a neural wearable and also support embedded ML readiness by developing electrophysiology interfaces, validating signal quality, and contributing to early product architecture.

QB3 NANOTECH RESEARCH LAB

Berkeley, CA

Electron-Beam Lithography and CNT Research Assistant

Aug 2025 – Present

- Researching low-cost, scalable lithography methods to reduce bottlenecks in chip production and increase accessibility.
- Developing chips that enable controlled carbon nanotube growth to enhance energy storage efficiency and power output.

TECHNICAL PROJECTS

CS61C PROJECT 3: RISC-V CPU DESIGN

Berkeley, CA

Hardware Design Project Partner

March 2026 – April 2026

Designed and debugged a two-stage pipelined RISC-V CPU in Logisim in a team of two for CS61C, implementing ALU, control logic, memory operations, branching, pipelining, and instruction execution across the RV32I ISA.

MIT BEAVERWORKS BWSI: BASICS OF ASICS

Cambridge, MA

Lead Student Engineer

Mar 2024 – Aug 2024

- Designed and fabricated a custom ASIC and PCB (NFC Business Card) using Verilog, OpenLane, and KiCAD, with final tapeout through TinyTapeout, XSchem, and SkyWater Tech Foundry.
- Developed “Brailliance”, a portable braille reader leveraging custom logic blocks optimized for accessibility and low-power use; reduced device cost by 90%. ([📺 Brailliance | The Affordable Braille Reader](#))
- Completed RTL design, simulation, and synthesis while collaborating with mentors to meet power and timing constraints.

AQUAPILOT - AI SPRINKLER SYSTEM

Irvine, CA

Founder and Developer

Feb 2024 – May 2025

- Engineered an AI-driven irrigation system integrating capacitive soil moisture sensors, DHT22 climate sensors, and a Raspberry Pi controller to autonomously regulate solenoid valves using supervised ML models.

TECHNICAL SKILLS

Languages: Python, Verilog, C, RISC-V, Java, C, SystemVerilog, JavaScript, HTML/CSS

Tools & Frameworks: Git, Valgrind, Docker, LTSpice, SolidWorks, Altium, GDB, KiCAD, Logisim, Makefile